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Exact solutions of Dirac equation for hydrogen atom using the linear combination of orthogonal Laguerre basis functions

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The associated Laguerre polynomials are very popular in the Schrödinger equation of nonrelativistic quantum mechanics. However, the situation is quite different in the case of the Dirac equation of relativistic quantum mechanics. The exact solutions of the Dirac equation for three-dimensional (3D) hydrogen atom have been obtained using the wavefunction expansion method. The exact solutions of the radial wavefunctions are found in the form of a linear combination of the orthogonal Laguerre basis function spinors. According to the equality of coefficients in Laguerre series on both sides of the equations, the exact eigenenergies and exact expressions of the corresponding eigenstates of the Dirac equation for 3D hydrogen atom are obtained, which are the same as the power series solutions. Additionally, a comparison with the results from Dirac and Schrödinger equations for 3D hydrogen atom is discussed. Due to the relativistic effects, the Dirac bispinor occurs, and there are the tiny differences of the results between Dirac and Schrödinger equations. For the Dirac states with $n_r = 0$, there is one set of the Dirac bispinor. However for the Dirac states with $n_r \neq 0$, there is a combination of two sets of the Dirac bispinor. For degenerate Dirac states, their exact expansion coefficients are commutative symmetric. These results show that the linear combination of the orthogonal Laguerre basis functions is the exact solutions of the Dirac equation for 3D hydrogen atom.

Keywords Dirac equation, 3D hydrogen atom, Exact solutions, Wavefunction expansion method, Orthogonal Laguerre basis functions

Since it was first proposed by Paul Dirac in 1928[1], the Dirac equation has attracted a great deal of attention among researchers in relativistic quantum theory[2–24]. It is well known that the Dirac equation plays a fundamental role in relativistic quantum mechanics since this equation successfully merges quantum mechanics with special relativity. It is a relativistically covariant linear first order differential equation in space and time for a four-component spinor wavefunction, and describes the quantum behavior of fermions at high speeds[25–29]. The Dirac equation of 3D hydrogen atom is of particular interest since the analytical solutions allow for a better understanding of physical phenomena and establish the necessary correspondence between relativistic effects and their nonrelativistic analogues.

Owing to four coupled partial differential equations, weak singularities at the point nucleus and the existence of the Dirac sea, there are still a number of difficulties for solving the Dirac equation stably and accurately[30]. The Dirac equation is more complex mathematically compared to the Schrödinger equation. It is usually claimed that the associated Laguerre polynomials were popularized by Schrödinger when creating the nonrelativistic quantum mechanics. However, the situation is quite different in the case of the Dirac equation of relativistic quantum mechanics. In contrast to the case of the Schrödinger equation, the associated Laguerre polynomials do not seem to be adopted by classical textbooks on relativistic quantum mechanics when solving the Dirac equations for the 3D hydrogen atom[31]. Since the orbital angular momentum quantum number is an integer while, of course, $\gamma = \sqrt{\kappa^2 - \alpha^2 Z^2}$ is not, only one associated Laguerre function is explicitly used in the solution of 3D hydrogen atom for the Schrödinger equation, but not for the Dirac equation. The exact solutions of the Dirac equation for 3D hydrogen atom make the description of many-electron atoms and molecules feasible. Thus, there is a need for systematic construction of the exact solutions of the Dirac equation for 3D hydrogen atom using the linear combination of the orthogonal Laguerre basis functions. In spite of the numerous studies realized, to the best of our knowledge, there have not been theoretical investigations on the exact solutions

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of the Dirac equation for 3D hydrogen atom using the wavefunction expansion method based on the linear combination of the orthogonal Laguerre basis functions. In this work, we present the exact solutions of the Dirac equation for 3D hydrogen atom using the wavefunction expansion method based on the linear combination of the orthogonal Laguerre basis functions.

This work is structured as follows. In sect. "Model and method", we present the exact solutions of the Dirac equation for 3D hydrogen atom using the wavefunction expansion method based on the linear combination of the orthogonal Laguerre basis functions. Next, in sect. "Results and discussion", the exact relativistic eigenenergies and the exact corresponding eigenstates are obtained and discussed. Summary is presented in sect. "Conclusions". The exact solutions of the nonrelativistic Schrödinger equation for 3D hydrogen atom are derived in detail in the Supplemental Material. Atomic units (a.u.) $\hbar = e = m_e = 1$ are used throughout the paper unless otherwise stated. The velocity of light in vacuum $c = 1/\alpha$ is the inverse of the Sommerfeld's fine-structure constant α, which is used as the CODATA 2010 value $1/\alpha = 137.035999074$[32].

Model and method

The Dirac Hamiltonian H for 3D hydrogen atom reads[33].

$$H = c\vec{\alpha} \cdot \vec{p} + (\beta - 1)c^2 + V(r) \tag{1}$$

where $\vec{p} = -i\nabla$ is the momentum operator, $\vec{\alpha}$ and β are 4×4 matrices of the Dirac operators. $V(r) = -Z/r$ is the Coulomb potential between the electron and the point nucleus, where r is the distance from the point nucleus to the electron and Z is the number of nuclear charges. For the hydrogen atom, $Z = 1$.

The Dirac equation of hydrogen atom is written as.

$$H\psi_{n\kappa m}(\vec{r}) = E\psi_{n\kappa m}(\vec{r}) \tag{2}$$

where E is the relativistic energy relative to the rest mass energy c^2 of the electron.

In a central Coulomb field of 3D hydrogen atom, the two-component spinor wavefunction $\psi_{n\kappa m}(\vec{r})$ in spherical coordinates is conveniently written as follows:

$$\psi_{n\kappa m}(\vec{r}) = \frac{1}{r}\begin{pmatrix} P_{n\kappa}(r)\chi_{\kappa m}(\theta,\phi) \\ iQ_{n\kappa}(r)\chi_{-\kappa m}(\theta,\phi) \end{pmatrix} \tag{3}$$

where i is the imaginary unit. For simplicity, the radial quantum number n_r is written as n omitting the subscript r, and $n = 0, 1, 2, \ldots$. The spin–orbit coupling quantum number $\kappa = \pm1, \pm2, \pm3, \ldots$ and $m = 0, \pm1, \pm2 \ldots$ is the projection of the angular momentum on the z-axis. $P_{n\kappa}(r)$ and $Q_{n\kappa}(r)$ represent the large and small components of the radial wavefunction, respectively. $\chi_{\kappa m}(\theta,\phi)$ and $\chi_{-\kappa m}(\theta,\phi)$ are the spherical harmonic spinors.

The components $P_{n\kappa}(r)$ and $Q_{n\kappa}(r)$ satisfy the common normalization condition, i.e.,

$$\int_0^\infty [|P_{n\kappa}(r)|^2 + |Q_{n\kappa}(r)|^2]dr = 1 \tag{4}$$

After a substitution of Eq. (3) into Eq. (2), it is convenient to separate the angle part into a set of coupled first-order ordinary differential equations of $P_{n\kappa}(r)$ and $Q_{n\kappa}(r)$ in matrix form for a given value of κ:

$$H\begin{pmatrix} P_{n\kappa}(r) \\ Q_{n\kappa}(r) \end{pmatrix} = E\begin{pmatrix} P_{n\kappa}(r) \\ Q_{n\kappa}(r) \end{pmatrix} \tag{5}$$

with the radial Hamiltonian of the matrix form.

$$H = \begin{pmatrix} V(r) & c(-\frac{\partial}{\partial r} + \frac{\kappa}{r}) \\ c(\frac{\partial}{\partial r} + \frac{\kappa}{r}) & V(r) - 2c^2 \end{pmatrix} \tag{6}$$

Then Eq. (5) can be recast in the following form.

$$-\frac{Z}{r}P_{n_r,\kappa}(r) + c[-\frac{\partial Q_{n_r,\kappa}(r)}{\partial r} + \frac{\kappa}{r}Q_{n_r,\kappa}(r)] = EP_{n_r,\kappa}(r) \tag{7a}$$

$$c[\frac{\partial P_{n_r,\kappa}(r)}{\partial r} + \frac{\kappa}{r}P_{n_r,\kappa}(r)] + (-\frac{Z}{r} - 2c^2)Q_{n_r,\kappa}(r) = EQ_{n_r,\kappa}(r) \tag{7b}$$

Combining the two limit cases of both radial components at $r \to 0$ and $r \to \infty$, the normalized complete orthogonal Laguerre basis functions $\varphi_{n\kappa}(r)$ are constructed by the following form[24].

$$\varphi_{n\kappa}(r) = [\frac{n!}{\Gamma(n + 2\gamma + 1)}]^{1/2}\zeta^{1/2}(\zeta r)^\gamma \exp(-\frac{\zeta r}{2})L_n^{2\gamma}(\zeta r) \tag{8}$$

with the orthogonality relation.

$$\int_0^\infty \varphi_{n'\kappa}(r)\varphi_{n\kappa}(r)dr = \delta_{n'n} \tag{9}$$

where $\{\varphi_{n\kappa}(r)\}$ is a normalized complete orthogonal set of basis functions for each value of $n = 0, 1, 2, 3 \ldots$, and a given value of $\kappa = \pm 1, \pm 2, \pm 3, \ldots$, in the domain $0 \leq r < \infty$. $\Gamma(x)$ is the Gamma function. $L_n^{2\gamma}(\zeta r)$ is the associated Laguerre polynomial with $\gamma = \sqrt{\kappa^2 - \alpha^2 Z^2}$ and $\zeta = 2\sqrt{-2E - \alpha^2 E^2}$. $\delta_{n'n}$ is the Kronecker delta function.

To solve Eqs. (7a) and (7b), the components $P_{n_r,\kappa}(r)$ and $Q_{n_r,\kappa}(r)$ are expanded in the linear combination of the orthogonal Laguerre basis functions, respectively.

$$P_{n_r,\kappa}(r) = \sum_{n=0}^{n_r} C_{n\kappa}^L \varphi_{n\kappa}(r) \tag{10a}$$

and

$$Q_{n_r,\kappa}(r) = \sum_{n=0}^{n_r} C_{n\kappa}^S \varphi_{n\kappa}(r) \tag{10b}$$

where $C_{n\kappa}^L$ and $C_{n\kappa}^S$ are the expansion coefficients, and the superscript L and S identify the large and small components of the Dirac spinor in a conventional way. The summation over n is truncated at the radial quantum number n_r chosen in an actual run.

Substituting Eqs. (10a) and (10b) into Eqs. (7a) and (7b), one easily obtains the following equations as.

$$-\frac{Z}{r}\sum_{n=0}^{n_r} C_{n\kappa}^L \varphi_{n\kappa}(r) + c(-\frac{\partial}{\partial r} + \frac{\kappa}{r})\sum_{n=0}^{n_r} C_{n\kappa}^S \varphi_{n\kappa}(r) = E\sum_{n=0}^{n_r} C_{n\kappa}^L \varphi_{n\kappa}(r) \tag{11a}$$

$$c(\frac{\partial}{\partial r} + \frac{\kappa}{r})\sum_{n=0}^{n_r} C_{n\kappa}^L \varphi_{n\kappa}(r) + (-\frac{Z}{r} - 2c^2)\sum_{n=0}^{n_r} C_{n\kappa}^S \varphi_{n\kappa}(r) = E\sum_{n=0}^{n_r} C_{n\kappa}^S \varphi_{n\kappa}(r) \tag{11b}$$

The derivative of associated Laguerre function is given by[34].

$$\frac{d}{dx}L_n^k(x) = -L_{n-1}^{(k+1)}(x) \tag{12}$$

Following Eq. (12), one obtains the first-order derivative with respect to r,

$$\begin{aligned}\frac{\partial \varphi_{n\kappa}(r)}{\partial r} &= \zeta^{3/2}[\frac{n!}{\Gamma(n+2\gamma+1)}]^{1/2}[\gamma(\zeta r)^{\gamma-1}\exp(-\frac{\zeta r}{2})L_n^{2\gamma}(\zeta r) \\ &\quad - \frac{1}{2}(\zeta r)^{\gamma}\exp(-\frac{\zeta r}{2})L_n^{2\gamma}(\zeta r) - (\zeta r)^{\gamma}\exp(-\frac{\zeta r}{2})L_{n-1}^{2\gamma+1}(\zeta r)]\end{aligned} \tag{13}$$

Substituting Eqs. (8) and (13) into Eqs. (11a) and (11b) and multiplying by $\zeta^{-\gamma-1/2}r^{-\gamma+1}\exp(\frac{\zeta r}{2})$ on both sides of the equations, one then obtains the following equations

$$\begin{aligned}&\sum_{n=0}^{n_r}[c(\kappa-\gamma)C_{n\kappa}^S - ZC_{n\kappa}^L][\frac{n!}{\Gamma(n+2\gamma+1)}]^{1/2}L_n^{2\gamma}(\zeta r) \\ &+\sum_{n=0}^{n_r}(\frac{1}{2}cC_{n\kappa}^S - \frac{E}{\zeta}C_{n\kappa}^L)[\frac{n!}{\Gamma(n+2\gamma+1)}]^{1/2}(\zeta r)L_n^{2\gamma}(\zeta r) \\ &+c\sum_{n=0}^{n_r}C_{n\kappa}^S[\frac{n!}{\Gamma(n+2\gamma+1)}]^{1/2}(\zeta r)L_{n-1}^{2\gamma+1}(\zeta r) = 0\end{aligned} \tag{14a}$$

$$\begin{aligned}&\sum_{n=0}^{n_r}[c(\kappa+\gamma)C_{n\kappa}^L - ZC_{n\kappa}^S][\frac{n!}{\Gamma(n+2\gamma+1)}]^{1/2}L_n^{2\gamma}(\zeta r) \\ &-\sum_{n=0}^{n_r}(\frac{1}{2}cC_{n\kappa}^L + \frac{E+2c^2}{\zeta}C_{n\kappa}^S)[\frac{n!}{\Gamma(n+2\gamma+1)}]^{1/2}(\zeta r)L_n^{2\gamma}(\zeta r) \\ &-c\sum_{n=0}^{n_r}C_{n\kappa}^L[\frac{n!}{\Gamma(n+2\gamma+1)}]^{1/2}(\zeta r)L_{n-1}^{2\gamma+1}(\zeta r) = 0\end{aligned} \tag{14b}$$

The recurrence relation of associated Laguerre functions is given by[34].

$$xL_n^k(x) = -(n+k)L_{n-1}^k(x) + (2n+k+1)L_n^k(x) - (n+1)L_{n+1}^k(x) \tag{15}$$

Following Eq. (15), the following recurrence relation is found.

$$(\zeta r)L_{n-1}^{2\gamma+1}(\zeta r) = -(n+2\gamma)L_{n-2}^{2\gamma+1}(\zeta r) + 2(n+\gamma)L_{n-1}^{2\gamma+1}(\zeta r) - nL_n^{2\gamma+1}(\zeta r) \tag{16}$$

The recurrence relations of associated Laguerre functions are given by[34].

$$L_n^k(x) = L_n^{k+1}(x) - L_{n-1}^{k+1}(x) \tag{17}$$

Following Eq. (17), the recurrence relation of associated Laguerre functions is given by[34].

$$L_n^{2\gamma}(\zeta r) = L_n^{2\gamma+1}(\zeta r) - L_{n-1}^{2\gamma+1}(\zeta r) \tag{18}$$

Substituting Eq. (18) into Eq. (15), the following recurrence relation is found

$$\begin{aligned}(\zeta r)L_n^{2\gamma}(\zeta r) = (n+2\gamma)L_{n-2}^{2\gamma+1}(\zeta r) - (3n+4\gamma+1)L_{n-1}^{2\gamma+1}(\zeta r)\\ + (3n+2\gamma+2)L_n^{2\gamma+1}(\zeta r) - (n+1)L_{n+1}^{2\gamma+1}(\zeta r)\end{aligned} \tag{19}$$

Substituting Eqs. (16) and (19) into Eqs. (14a) and (14b), one easily obtains the following equations as

$$\begin{aligned}&-\sum_{n=0}^{n_r}(n+2\gamma)(\frac{1}{2}cC_{n\kappa}^S + \frac{E}{\zeta}C_{n\kappa}^L)[\frac{n!}{\Gamma(n+2\gamma+1)}]^{1/2}L_{n-2}^{2\gamma+1}(\zeta r)\\ &+\sum_{n=0}^{n_r}\{c(-\kappa+\gamma+\frac{1}{2}n-\frac{1}{2})C_{n\kappa}^S + [Z+\frac{E}{\zeta}(3n+4\gamma+1)]C_{n\kappa}^L\}[\frac{n!}{\Gamma(n+2\gamma+1)}]^{1/2}L_{n-1}^{2\gamma+1}(\zeta r)\\ &+\sum_{n=0}^{n_r}\{c(\kappa+\frac{1}{2}n+1)C_{n\kappa}^S - [Z+\frac{E}{\zeta}(3n+2\gamma+2)]C_{n\kappa}^L\}[\frac{n!}{\Gamma(n+2\gamma+1)}]^{1/2}L_n^{2\gamma+1}(\zeta r)\\ &-\sum_{n=0}^{n_r}(n+1)(\frac{1}{2}cC_{n\kappa}^S - \frac{E}{\zeta}C_{n\kappa}^L)[\frac{n!}{\Gamma(n+2\gamma+1)}]^{1/2}L_{n+1}^{2\gamma+1}(\zeta r) = 0\end{aligned} \tag{20a}$$

$$\begin{aligned}&\sum_{n=0}^{n_r}(n+2\gamma)(\frac{1}{2}cC_{n\kappa}^L - \frac{E+2c^2}{\zeta}C_{n\kappa}^S)[\frac{n!}{\Gamma(n+2\gamma+1)}]^{1/2}L_{n-2}^{2\gamma+1}(\zeta r)\\ &+\sum_{n=0}^{n_r}\{c(-\kappa-\gamma-\frac{1}{2}n+\frac{1}{2})C_{n\kappa}^L + [Z+\frac{E+2c^2}{\zeta}(3n+4\gamma+1)]C_{n\kappa}^S\}[\frac{n!}{\Gamma(n+2\gamma+1)}]^{1/2}L_{n-1}^{2\gamma+1}(\zeta r)\\ &+\sum_{n=0}^{n_r}\{c(\kappa-\frac{1}{2}n-1)C_{n\kappa}^L - [Z+\frac{E+2c^2}{\zeta}(3n+2\gamma+2)]C_{n\kappa}^S\}[\frac{n!}{\Gamma(n+2\gamma+1)}]^{1/2}L_n^{2\gamma+1}(\zeta r)\\ &+\sum_{n=0}^{n_r}(n+1)(\frac{1}{2}cC_{n\kappa}^L + \frac{E+2c^2}{\zeta}C_{n\kappa}^S)[\frac{n!}{\Gamma(n+2\gamma+1)}]^{1/2}L_{n+1}^{2\gamma+1}(\zeta r) = 0\end{aligned} \tag{20b}$$

After the associated Laguerre functions are expanded in the Laguerre series of $(2\gamma+1)$ order, we obtain in the Laguerre series of the associated Laguerre functions of $(2\gamma+1)$ order on both sides of Eqs. (20a) and (20b). According to the equality of coefficients in the same Laguerre order on both sides of the equation, the equations of the exact relativistic eigenenergies and the exact expansion coefficients of the corresponding eigenstates will be obtained.

Setting the expansion coefficients of $L_{n_r+1}^{2\gamma+1}(\zeta r)$, $L_{n_r}^{2\gamma+1}(\zeta r)$, $L_{n_r-1}^{2\gamma+1}(\zeta r)$ and $L_{n_r-2-\mu}^{2\gamma+1}(\zeta r)$ to zero in Eq. (20a), we have the following equations.

$$\frac{1}{2}cC_{n_r,\kappa}^S - \frac{E}{\zeta}C_{n_r,\kappa}^L = 0 \tag{21a}$$

$$\begin{aligned}&-n_r(\frac{1}{2}cC_{n_r-1,\kappa}^S - \frac{E}{\zeta}C_{n_r-1,\kappa}^L)[\frac{(n_r-1)!}{\Gamma(n_r+2\gamma)}]^{1/2}\\ &+\{c(\kappa+\frac{1}{2}n_r+1)C_{n_r,\kappa}^S - [Z+\frac{E}{\zeta}(3n_r+2\gamma+2)]C_{n_r,\kappa}^L\}[\frac{n_r!}{\Gamma(n_r+2\gamma+1)}]^{1/2} = 0\end{aligned} \tag{21b}$$

$$
\begin{aligned}
&-(n_r-1)(\frac{1}{2}cC^S_{n_r-2,\kappa}-\frac{E}{\zeta}C^L_{n_r-2,\kappa})[\frac{(n_r-2)!}{\Gamma(n_r+2\gamma-1)}]^{1/2}\\
&+\{c(\kappa+\frac{1}{2}n_r+\frac{1}{2})C^S_{n_r-1,\kappa}-[Z+\frac{E}{\zeta}(3n_r+2\gamma-1)]C^L_{n_r-1,\kappa}\}[\frac{(n_r-1)!}{\Gamma(n_r+2\gamma)}]^{1/2}\\
&+\{c(-\kappa+\gamma+\frac{1}{2}n_r-\frac{1}{2})C^S_{n_r,\kappa}+[Z+\frac{E}{\zeta}(3n_r+4\gamma+1)]C^L_{n_r,\kappa}\}[\frac{n_r!}{\Gamma(n_r+2\gamma+1)}]^{1/2}=0
\end{aligned}
\tag{21c}
$$

$$
\begin{aligned}
&-(n_r-2-\mu)(\frac{1}{2}cC^S_{n_r-3-\mu,\kappa}-\frac{E}{\zeta}C^L_{n_r-3-\mu,\kappa})[\frac{(n_r-3-\mu)!}{\Gamma(n_r+2\gamma-2-\mu)}]^{1/2}\\
&+\{c(\kappa+\frac{1}{2}n_r-\frac{1}{2}\mu)C^S_{n_r-2-\mu,\kappa}-[Z+\frac{E}{\zeta}(3n_r+2\gamma-4-3\mu)]C^L_{n_r-2-\mu,\kappa}\}\\
&[\frac{(n_r-2-\mu)!}{\Gamma(n_r+2\gamma-1-\mu)}]^{1/2}+\{c(-\kappa+\gamma+\frac{1}{2}n_r-1-\frac{1}{2}\mu)C^S_{n_r-1-\mu,\kappa}\\
&+[Z+\frac{E}{\zeta}(3n_r+4\gamma-2-3\mu)]C^L_{n_r-1-\mu,\kappa}\}[\frac{(n_r-1-\mu)!}{\Gamma(n_r+2\gamma-\mu)}]^{1/2}\\
&-(n_r+2\gamma-\mu)(\frac{1}{2}cC^S_{n_r-\mu,\kappa}+\frac{E}{\zeta}C^L_{n_r-\mu,\kappa})[\frac{(n_r-\mu)!}{\Gamma(n_r+2\gamma+1-\mu)}]^{1/2}=0
\end{aligned}
\tag{21d}
$$

Setting the expansion coefficients of $L^{2\gamma+1}_{n_r+1}(\zeta r)$, $L^{2\gamma+1}_{n_r}(\zeta r)$, $L^{2\gamma+1}_{n_r-1}(\zeta r)$ and $L^{2\gamma+1}_{n_r-2-\mu}(\zeta r)$ to zero in Eq. (20b), we have the following equations.

$$
\frac{1}{2}cC^L_{n_r,\kappa}+\frac{E+2c^2}{\zeta}C^S_{n_r,\kappa}=0
\tag{22a}
$$

$$
\begin{aligned}
&n_r(\frac{1}{2}cC^L_{n_r-1,\kappa}+\frac{E+2c^2}{\zeta}C^S_{n_r-1,\kappa})[\frac{(n_r-1)!}{\Gamma(n_r+2\gamma)}]^{1/2}\\
&+\{c(\kappa-\frac{1}{2}n_r-1)C^L_{n_r,\kappa}-[Z+\frac{E+2c^2}{\zeta}(3n_r+2\gamma+2)]C^S_{n_r,\kappa}\}[\frac{n_r!}{\Gamma(n_r+2\gamma+1)}]^{1/2}=0
\end{aligned}
\tag{22b}
$$

$$
\begin{aligned}
&(n_r-1)(\frac{1}{2}cC^L_{n_r-2,\kappa}+\frac{E+2c^2}{\zeta}C^S_{n_r-2,\kappa})[\frac{(n_r-2)!}{\Gamma(n_r+2\gamma-1)}]^{1/2}\\
&+\{c(\kappa-\frac{1}{2}n_r-\frac{1}{2})C^L_{n_r-1,\kappa}-[Z+\frac{E+2c^2}{\zeta}(3n_r+2\gamma-1)]C^S_{n_r-1,\kappa}\}[\frac{(n_r-1)!}{\Gamma(n_r+2\gamma)}]^{1/2}\\
&+\{c(-\kappa-\gamma-\frac{1}{2}n_r+\frac{1}{2})C^L_{n_r,\kappa}+[Z+\frac{E+2c^2}{\zeta}(3n_r+4\gamma+1)]C^S_{n_r,\kappa}\}[\frac{n_r!}{\Gamma(n_r+2\gamma+1)}]^{1/2}=0
\end{aligned}
\tag{22c}
$$

$$
\begin{aligned}
&(n_r-2-\mu)(\frac{1}{2}cC^L_{n_r-3-\mu,\kappa}+\frac{E+2c^2}{\zeta}C^S_{n_r-3-\mu,\kappa})[\frac{(n_r-3-\mu)!}{\Gamma(n_r+2\gamma-2-\mu)}]^{1/2}\\
&+\{c(\kappa-\frac{1}{2}n_r+\frac{1}{2}\mu)C^L_{n_r-2-\mu,\kappa}-[Z+\frac{E+2c^2}{\zeta}(3n_r+2\gamma-4-3\mu)]C^S_{n_r-2-\mu,\kappa}\}\\
&[\frac{(n_r-2-\mu)!}{\Gamma(n_r+2\gamma-1-\mu)}]^{1/2}+\{c(-\kappa-\gamma-\frac{1}{2}n_r+1+\frac{1}{2}\mu)C^L_{n_r-1-\mu,\kappa}\\
&+[Z+\frac{E+2c^2}{\zeta}(3n_r+4\gamma-2-3\mu)]C^S_{n_r-1-\mu,\kappa}\}[\frac{(n_r-1-\mu)!}{\Gamma(n_r+2\gamma-\mu)}]^{1/2}\\
&+(n_r+2\gamma-\mu)(\frac{1}{2}cC^L_{n_r-\mu,\kappa}-\frac{E+2c^2}{\zeta}C^S_{n_r-\mu,\kappa})[\frac{(n_r-\mu)!}{\Gamma(n_r+2\gamma+1-\mu)}]^{1/2}=0
\end{aligned}
\tag{22d}
$$

where $\mu=0,1,2,\ldots$ is an integer greater than or equal to zero.

Multiply Eq. (21b) by $(E+2c^2)$, multiply Eq. (22b) by $\frac{1}{2}c\zeta$, and add the two equations, we obtain the following relation.

$$
\frac{C^L_{n_r,\kappa}}{C^S_{n_r,\kappa}}=\frac{c^2\zeta(\kappa-n_r-\gamma)+2ZE}{2cE(\kappa+n_r+\gamma)+Zc\zeta}
\tag{23}
$$

Substituting $\zeta=2\sqrt{-2E-\alpha^2E^2}$, Eq. (23) combined with Eq. (21a) yields the eigenenergy equation as.

$$
[(n_r+\gamma)^2+\alpha^2Z^2]E^2+2[c^2(n_r+\gamma)^2+Z^2]E+Z^2c^2=0
\tag{24}
$$

By solving the above equation, the exact relativistic eigenenergies $E_{n_r,\kappa}$ are obtained as.

$$E_{n_r,\kappa} = c^2(\pm\frac{1}{\sqrt{1+\frac{\alpha^2 Z^2}{(n_r+\sqrt{\kappa^2-\alpha^2 Z^2})^2}}} - 1) \tag{25}$$

Only the root of the positive sign can be accepted, thus the exact relativistic eigenenergies $E_{n_r,\kappa}$ are obtained as.

$$E_{n_r,\kappa} = c^2(\frac{1}{\sqrt{1+\frac{\alpha^2 Z^2}{(n_r+\sqrt{\kappa^2-\alpha^2 Z^2})^2}}} - 1) \tag{26}$$

in which the radial quantum number $n_r = 0, 1, 2, \ldots,$ and the spin–orbit coupling quantum number $\kappa = \pm 1, \pm 2, \pm 3, \ldots$.

When $n_r = 0$, by solving Eq. (21a), one easily obtains the expansion coefficient relationship as.

$$\frac{C^L_{n_r,\kappa}}{C^S_{n_r,\kappa}} = \frac{c\zeta}{2E} \tag{27}$$

After normalizing the wave functions, the exact expansion coefficients $C^L_{n_r,\kappa}$ and $C^S_{n_r,\kappa}$ are obtained as.

$$C^L_{n_r,\kappa} = \frac{1}{\sqrt{1+(\frac{2E}{c\zeta})^2}} \tag{28a}$$

$$C^S_{n_r,\kappa} = \frac{\frac{2E}{c\zeta}}{\sqrt{1+(\frac{2E}{c\zeta})^2}} \tag{28b}$$

Multiply Eq. (21c) by $(E + 2c^2)$, multiply Eq. (22c) by $\frac{1}{2}c\zeta$, and add the two equations, we obtain the following relation.

$$\begin{aligned}
&\{[\frac{1}{2}c^2\zeta(\kappa+n_r+\gamma-1) - Z(E+2c^2)]C^L_{n_r-1,\kappa} \\
&+[c(E+2c^2)(\kappa-n_r-\gamma+1) - \frac{1}{2}Zc\zeta]C^S_{n_r-1,\kappa}\}[\frac{(n_r-1)!}{\Gamma(n_r+2\gamma)}]^{1/2} \\
&+\{[-\frac{1}{2}c^2\zeta(\kappa+2n_r+3\gamma) + Z(E+2c^2)]C^L_{n_r,\kappa} \\
&+[-c(E+2c^2)(\kappa-2n_r-3\gamma) + \frac{1}{2}Zc\zeta]C^S_{n_r,\kappa}\}[\frac{n_r!}{\Gamma(n_r+2\gamma+1)}]^{1/2} = 0
\end{aligned} \tag{29}$$

When $n_r \neq 0$, by solving Eqs. (21a), (21b) and (29) and normalizing the wave functions, the exact expansion coefficients $C^L_{n_r,\kappa}$, $C^S_{n_r,\kappa}$, $C^L_{n_r-1,\kappa}$ and $C^S_{n_r-1,\kappa}$ are obtained as.

$$C^L_{n_r,\kappa} = \frac{1}{\sqrt{1+(\frac{2E}{c\zeta})^2+(a^L_{n_r-1,\kappa})^2+(a^S_{n_r-1,\kappa})^2}} \tag{30a}$$

$$C^S_{n_r,\kappa} = \frac{\frac{2E}{c\zeta}}{\sqrt{1+(\frac{2E}{c\zeta})^2+(a^L_{n_r-1,\kappa})^2+(a^S_{n_r-1,\kappa})^2}} \tag{30b}$$

$$C^L_{n_r-1,\kappa} = \frac{a^L_{n_r-1,\kappa}}{\sqrt{1+(\frac{2E}{c\zeta})^2+(a^L_{n_r-1,\kappa})^2+(a^S_{n_r-1,\kappa})^2}} \tag{30c}$$

$$C^S_{n_r-1,\kappa} = \frac{a^S_{n_r-1,\kappa}}{\sqrt{1+(\frac{2E}{c\zeta})^2+(a^L_{n_r-1,\kappa})^2+(a^S_{n_r-1,\kappa})^2}} \tag{30d}$$

where

$$\begin{aligned}
a^L_{n_r-1,\kappa} = &-[\frac{1}{n_r(n_r+2\gamma)}]^{1/2}\{[2E(\kappa-n_r-\gamma) - Z\zeta][(E+2c^2)(\kappa-n_r-\gamma+1) - \frac{1}{2}Z\zeta] \\
&-\frac{1}{2}c^2\zeta^2 n_r(2n_r+3\gamma) + Z\zeta(E+c^2)n_r\}/[\frac{1}{2}c^2\zeta^2(n_r+\gamma-1) - Z\zeta(E+c^2)]
\end{aligned} \tag{31a}$$

$$a^S_{n_r-1,\kappa} = \frac{1}{c}[\frac{1}{n_r(n_r+2\gamma)}]^{1/2}\{[\frac{1}{2}c^2\zeta(\kappa+n_r+\gamma-1) - Z(E+2c^2)][2E(\kappa-n_r-\gamma) - Z\zeta] + c^2E\zeta n_r(2n_r+3\gamma) - 2ZE(E+c^2)n_r\}/[\frac{1}{2}c^2\zeta^2(n_r+\gamma-1) - Z\zeta(E+c^2)] \tag{31b}$$

$$\zeta = 2\sqrt{-2E-\alpha^2E^2} \tag{31c}$$

$$E = c^2(\frac{1}{\sqrt{1+\frac{\alpha^2Z^2}{(n_r+\sqrt{\kappa^2-\alpha^2Z^2})^2}}} - 1) \tag{31d}$$

Multiply Eq. (21d) by $(E+2c^2)$, multiply Eq. (22d) by $\frac{1}{2}c\zeta$, and add the two equations, we obtain the following relation

$$\begin{aligned}&\{[\frac{1}{2}c^2\zeta(\kappa+n_r+\gamma-2-\mu) - Z(E+2c^2)]C^L_{n_r-2-\mu,\kappa}\\&+[c(E+2c^2)(\kappa-n_r-\gamma+\mu+2) - \frac{1}{2}Zc\zeta]C^S_{n_r-2-\mu,\kappa}\}[\frac{(n_r-2-\mu)!}{\Gamma(n_r+2\gamma-1-\mu)}]^{1/2}\\&+\{[-\frac{1}{2}c^2\zeta(\kappa+2n_r+3\gamma-2-2\mu) + Z(E+2c^2)]C^L_{n_r-1-\mu,\kappa}\\&+[-c(E+2c^2)(\kappa-2n_r-3\gamma+2+2\mu) + \frac{1}{2}Zc\zeta]C^S_{n_r-1-\mu,\kappa}\}[\frac{(n_r-1-\mu)!}{\Gamma(n_r+2\gamma-\mu)}]^{1/2}\\&+(n_r+2\gamma-\mu)[\frac{1}{2}c^2\zeta C^L_{n_r-\mu,\kappa} - c(E+2c^2)C^S_{n_r-\mu,\kappa}][\frac{(n_r-\mu)!}{\Gamma(n_r+2\gamma+1-\mu)}]^{1/2} = 0\end{aligned} \tag{32}$$

By solving Eqs. (21c) and (32), where μ in Eq. (32) is set to $\mu=0$, the exact expansion coefficients $C^L_{n_r-2,\kappa}$ and $C^S_{n_r-2,\kappa}$ are obtained as.

$$C^L_{n_r-2,\kappa} = 0 \tag{33a}$$

$$C^S_{n_r-2,\kappa} = 0 \tag{33b}$$

By solving Eqs. (21d) and (32), where μ in Eq. (21d) is set to $\mu=0$ and μ in Eq. (32) is set to $\mu=1$, the exact expansion coefficients $C^L_{n_r-3,\kappa}$ and $C^S_{n_r-3,\kappa}$ are obtained as.

$$C^L_{n_r-3,\kappa} = 0 \tag{34a}$$

$$C^S_{n_r-3,\kappa} = 0 \tag{34b}$$

By solving Eqs. (21d) and (32), where μ in Eq. (21d) is set to $\mu=1$, and μ in Eq. (32) is set to $\mu=2$, the exact expansion coefficients $C^L_{n_r-4,\kappa}$ and $C^S_{n_r-4,\kappa}$ are obtained as.

$$C^L_{n_r-4,\kappa} = 0 \tag{35a}$$

$$C^S_{n_r-4,\kappa} = 0 \tag{35b}$$

By solving Eqs. (21d) and (32), where μ in Eq. (21d) is set to $\mu+2$, and μ in Eq. (32) is set to $\mu+3$, the exact expansion coefficients $C^L_{n_r-5-\mu,\kappa}$ and $C^S_{n_r-5-\mu,\kappa}$ are obtained as.

$$C^L_{n_r-5-\mu,\kappa} = 0 \tag{36a}$$

$$C^S_{n_r-5-\mu,\kappa} = 0 \tag{36b}$$

where $\mu = 0, 1, 2, \ldots$ is an integer greater than or equal to zero.

Results and discussion

The relativistic eigenenergies $E_{n_r,\kappa}$, the expansion coefficients $C^L_{n\kappa}$ and $C^S_{n\kappa}$ calculated from the Dirac equation for low-lying bound-states are presented in Table 1. When the relativistic effects are considered, the behaviour of an electron in the Coulomb potential must be described by the Dirac equation. In a central Coulomb field of 3D hydrogen atom, neither L nor S commute with the Dirac Hamiltonian H, while the total angular momentum operator J and spin–orbit coupling operator $\kappa = -\beta(\sigma \cdot L + 1)$ commute with H, where L and S are the orbital and spin angular momentum operator, respectively[15]. The complete conservative quantity set of an electron in the central field can be taken to be (H, K, J^2, J_z), thus the Dirac spinors can be classified according to the principal quantum number n, the spin–orbit coupling quantum number κ and the total angular momentum quantum number j. As usual, here one uses the standard spectroscopic notation (nl_j) referring to the large component of the Dirac spinor, where $j = |\kappa| - \frac{1}{2}$, $l = j + \frac{1}{2}\mathrm{sgn}\kappa$, and the principal quantum number $n = n_r + |\kappa| = 1, 2, 3, \ldots$. In Eq. (25), $E_{n\kappa} = -0.500007$ when $n_r = 0$ and $\kappa = -1$, whereas

$\vert nl_j\rangle$	n_r	κ	$E_{n_r,\kappa}$	$C^L_{n\kappa}$	$C^S_{n\kappa}$
$1s_{1/2}$	0	−1	−0.5000066565940845	$C^L_{0,-1}=0.999993$	$C^S_{0,-1}=-0.0036487$
$2s_{1/2}$	1	−1	−0.125002080189065	$C^L_{0,-1}=-0.499998$ $C^L_{1,-1}=0.866025$	$C^S_{0,-1}=-0.000912174C^S_{1,-1}=-0.00157994$
$2p_{1/2}$	1	+1	−0.125002080189065	$C^L_{0,+1}=-0.866025$ $C^L_{1,+1}=0.499998$	$C^S_{0,+1}=-0.00157994$ $C^S_{1,+1}=-0.000912174$
$2p_{3/2}$	0	−2	−0.12500041602688944	$C^L_{0,-2}=0.999998$	$C^S_{0,-2}=-0.00182434$
$3s_{1/2}$	2	−1	−0.05555629517642554	$C^L_{0,-1}=0$ $C^L_{1,-1}=-0.577349$ $C^L_{2,-1}=0.816497$	$C^S_{0,-1}=0$ $C^S_{1,-1}=-0.000702192C^S_{2,-1}=-0.000993051$
$3p_{1/2}$	2	+1	−0.05555629517642554	$C^L_{0,+1}=0$ $C^L_{1,+1}=-0.816497$ $C^L_{2,+1}=0.577349$	$C^S_{0,+1}=0$ $C^S_{1,+1}=-0.000993051$ $C^S_{2,+1}=-0.000702192$
$3p_{3/2}$	1	−2	−0.05555580208940604	$C^L_{0,-2}=-0.408247C^L_{1,-2}=0.912871$	$C^S_{0,-2}=-0.000496522C^S_{1,-2}=-0.00111026$
$3d_{3/2}$	1	+2	−0.05555580208940604	$C^L_{0,+2}=-0.912871$ $C^L_{1,+2}=0.408247$	$C^S_{0,+2}=-0.00111026$ $C^S_{1,+2}=-0.000496522$
$3d_{5/2}$	0	−3	−0.05555563773262587	$C^L_{0,-3}=0.999999$	$C^S_{0,-3}=-0.00121623$

Table 1. The relativistic eigenenergies $E_{n_r,\kappa}$ (a.u.) and the expansion coefficients $C^L_{n\kappa}$ and $C^S_{n\kappa}$ calculated from the Dirac equation for low-lying bound-states.

$E_{n\kappa}=-37557.2$ when $n_r=0$ and $\kappa=+1$. Thus only $\kappa=-1$ can be accepted when $n_r=0$. It is concluded that $n_r=0,1,2,\ldots$ if $\kappa<0$ and $n_r=1,2,3,\ldots$ if $\kappa>0$. The relativistic eigenenergies depend the principal quantum number n and the total angular momentum j. Except for the states with $n_r=0$ for which there are no solutions with positive κ, the Dirac states with the same n_r and j but different are degenerate with opposite values of κ, i.e., have the same energy. For instances, the state $1s_{1/2}$ is a singlet, and the states $2s_{1/2}$ and $2p_{1/2}$ are degenerate. However the degeneracy between the states with the same n but different j is removed, resulting in a fine structure splitting of the energy spectra of 3D hydrogen atom. Relativistic effects can be divided into scalar relativistic effects and spin–orbit coupling effects, both leading to the well-known fine structure of the spectra of the hydrogen atom. For the atoms with small Z (for instance, hydrogen atom), the velocity of the electron is much less than the speed of light, the relativistic effects have relatively weak effects on the fine structure.

It can be seen that for all the results in Table 1, the exact relativistic eigenenergies are the same as the power series solutions of Dirac equation for 3D hydrogen atom [see Eq. (26)]. These indicate that the linear combination of the orthogonal Laguerre basis functions is expressed as the exact solution of the Dirac equation for 3D hydrogen atom, which make the description of many-electron atoms and molecules feasible. The degree of the associated Laguerre function must be a nonnegative integer. Since the orbital angular momentum quantum number is a nonnegative integer while, of course, $\gamma=\sqrt{\kappa^2-\alpha^2Z^2}$ is not, only one associated Laguerre function is explicitly used in the solution of 3D hydrogen atom for the Schrödinger equation, but not for the Dirac equation. It is also shown in Table 1 that the expansion coefficients of the large components of the Dirac spinor are much larger than those of the small components. When the relativistic effects are taken into account, the coupling of the large and small spinor components of the Dirac equation occurs, which leads to the coefficient of the small component is very small. For the Dirac states with $n_r=0$, there is one set of the Dirac bispinor. However for the Dirac states with $n_r\neq 0$, there is a combination of two sets of the Dirac bispinor. For degenerate Dirac states, their expansion coefficients are commutative symmetric. The exact solutions of the Dirac equation for 3D hydrogen atom result from a combination of three aspects to our method. First, the associated Laguerre polynomials are very popular in quantum mechanics, and a complete set of orthogonal Laguerre basis functions has been constructed explicitly for relativistic calculations. The radial prefactor r^γ with $\gamma=\sqrt{\kappa^2-\alpha^2Z^2}$ gives a realistic behavior of the wavefunction close to a point nucleus. Second, the associated Laguerre basis functions have rather rapid convergence[35]. Third, the fully explicit expressions of the Dirac Hamiltonian can be derived from the method of the generating function of the associated Laguerre polynomials. Using the rules for fractional derivatives, γ may be taken the $(\gamma-1)th$ derivative.

In order to compare with nonrelativistic calculations, the nonrelativistic eigenenergies $E_{n_r,l}$ and the expansion coefficients C_{nl} calculated from the Schrödinger equation for low-lying bound-states are also shown in Table 2 [see the Supplemental Material]. In the nonrelativistic limit, the Dirac equation reduces to the Schrödinger equation for nonrelativistic quantum mechanics. In the nonrelativistic theory of the hydrogen atom, the energy levels depend only upon the principal quantum number n. Compared with the results from Table 1, we find that the tiny differences occur due to the relativistic effects. The relativistic corrections are very small in comparison with the corresponding nonrelativistic eigenenergies. When the relativistic effects are ignored, the small components go to zero and the large components thus become the solutions of the corresponding nonrelativistic Schrödinger equation. In nonrelativistic quantum mechanics, the spin and orbital degrees of freedom are independent. If the electron spin is taken into account, each solution of the Schrödinger equation is doubled to two spin states, $\psi_{nlm}(\vec{r})\chi_{1/2}$ and $\psi_{nlm}(\vec{r})\chi_{-1/2}$, where $\chi_{1/2}$ and $\chi_{-1/2}$ are the so-called spin states which correspond to the 1/2 and -1/2 values of the spin quantum number m_s, respectively.

$\|nl\rangle$	n_r		$E_{n_r,l}$	C_{nl}
1 s	0	0	−0.5	$C_{0,0}=1$
2 s	1	0	−0.125	$C_{0,0}=-0.5$ $C_{1,0}=0.866025$
2p	0	1	−0.125	$C_{0,1}=1$
3 s	2	0	−0.0555555555555555	$C_{0,0}=0$ $C_{1,0}=-0.57735$ $C_{2,0}=0.816497$
3p	1	1	−0.0555555555555555	$C_{0,1}=-0.408248$ $C_{1,1}=0.912871$
3d	0	2	−0.0555555555555555	$C_{0,2}=1$

Table 2. The nonrelativistic eigenenergies $E_{n_r,l}$ (a.u.) and the expansion coefficients C_{nl} calculated from the Schrödinger equation for low-lying bound-states.

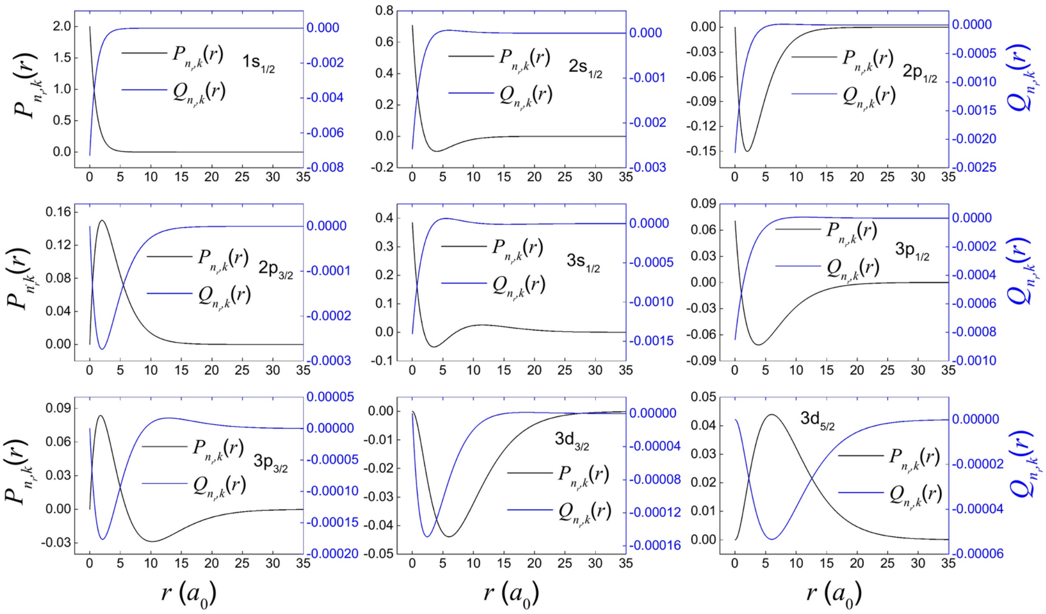

Fig. 1. The large and small radial components $P_{n_r,\kappa}(r)/r$ (black, left scale) and $Q_{n_r,\kappa}(r)/r$ (blue, right scale) of relativistic Dirac states versus r for low-lying bound-states.

To get a detailed insight into the behavior of the components, Fig. 1 plots the large and small radial components $P_{n_r,\kappa}(r)/r$ (black, left scale) and $Q_{n_r,\kappa}(r)/r$ (blue, right scale) of relativistic Dirac states versus r for low-lying bound-states. The Dirac equation describes electrons with 4-component spinors, rather than scalar wavefunctions. The amplitude of the large component $P_{n_r,\kappa}(r)/r$ is much larger than that of the small component $Q_{n_r,\kappa}(r)/r$. For $\kappa<0$, the role of the two components is reversed for each other, while for $\kappa>0$, the role of the two components is consistent for each other. For $|\kappa|=1$, the Dirac spinors exhibit slight singularity at the origin but can still be normalized, making them acceptable. The large component of the wavefunction, $P_{n_r,\kappa}(r)/r$, has always $n-l-1$ nodes, while the $Q_{n_r,\kappa}(r)/r$ component has $n-l-1$ nodes if $\kappa<0$, and $n-1$ nodes if $\kappa>0$. However the $Q_{n_r,\kappa}(r)/r$ component of $3d_{3/2}$ state with $\kappa=2$ is an exception, which has only one node, not two nodes.

In order to compare with nonrelativistic calculations, the radial wavefunctions $R_{n_r,l}(r)/r$ of nonrelativistic Schrödinger states versus r for low-lying bound-states are also shown in Fig. 2. In nonrelativistic quantum mechanics, both spin and orbital motions are independent. In the absence of spin–orbit coupling, the small components go to zero and the large components thus become the solutions of the corresponding nonrelativistic Schrödinger equation. Note that all states of Schrödinger equation are consistent with the large components of the Dirac spinor, only with tiny differences. The electron spin projection is conserved, and the spinor has two spin states, $\chi_{1/2}$ and $\chi_{-1/2}$. The radial wave functions $R_{n_r,l}(r)/r$ have $n-l-1$ nodes.

Conclusions

In this work, we have obtained the exact solutions of the Dirac equation for 3D hydrogen atom using the wavefunction expansion method based on the linear combination of the orthogonal Laguerre basis functions. According to the equality of coefficients in Laguerre series on both sides of the equations, the exact eigenenergies

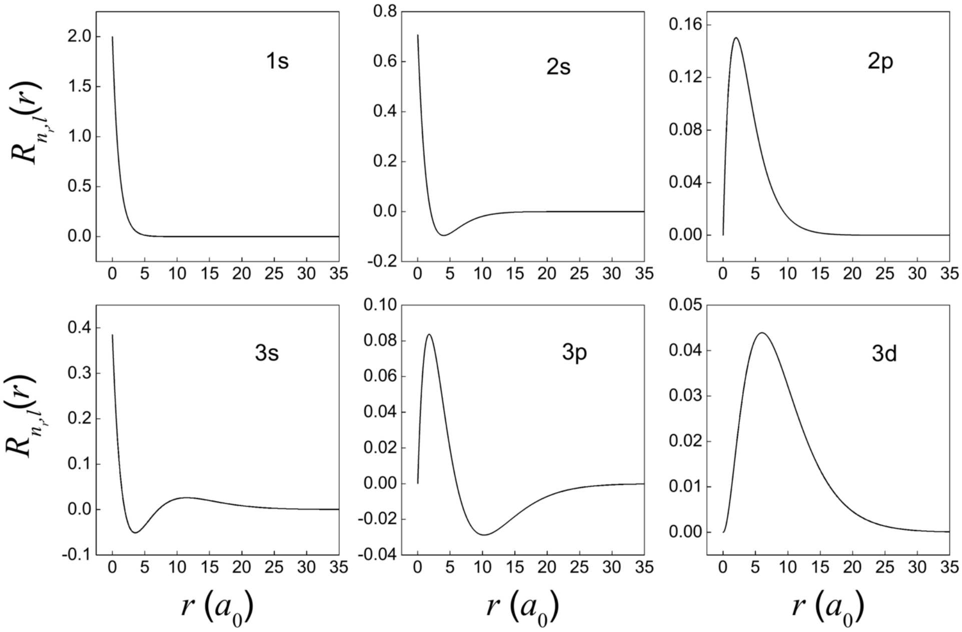

Fig. 2. The radial wavefunctions $R_{n_r,l}(r)/r$ of nonrelativistic Schrödinger states versus r for low-lying bound-states.

and exact expressions of the corresponding eigenstates of the Dirac equation for 3D hydrogen atom are obtained. The exact relativistic eigenenergies are the same as the power series solutions of Dirac equation for 3D hydrogen atom. A comparison with the results from the Schrödinger equation is discussed. The tiny differences of the results between Dirac and Schrödinger equations occur due to the relativistic effects. These results show that the linear combination of the orthogonal Laguerre basis functions is the exact solutions of the Dirac equation for 3D hydrogen atom.

Data availability

Data is provided within the manuscript or supplementary information files.

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Author contributions
S.W. wrote the main manuscript text, prepared all figures and reviewed the manuscript.

Declarations

Competing interests
The authors declare no competing interests.

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